Lecture 3

Direction Fields

We may be considering an IVP that is guaranteed a solution by the E-U Theorem, however, the E-U Theorem does not prescribe a technique to determine what the solution is. In many circumstances, there is no technique to determine the solution. In this lecture we will explore one avenue of understanding the properties and nature of the solution without the need to determine it explicitly.

A useful strategy for a 1st order differential equation is to sketch a direction field for the equation.

When a 1st order equation is written in the form

slope of y

$$\boxed{\frac{dy}{dx} = f(x, y)}$$

expression for slope at (x, y)

it provides a formula for the slope of the solution $y(x)$ at any (x, y).

Definition: A **direction field** is a plot of short line segments (tangent line segments) drawn at various points in the xy-plane showing the slope of a solution curve at that point.

A direction field provides a visualization of the "flow of solutions" and allows the drawing of a particular solution at a set of initial conditions.

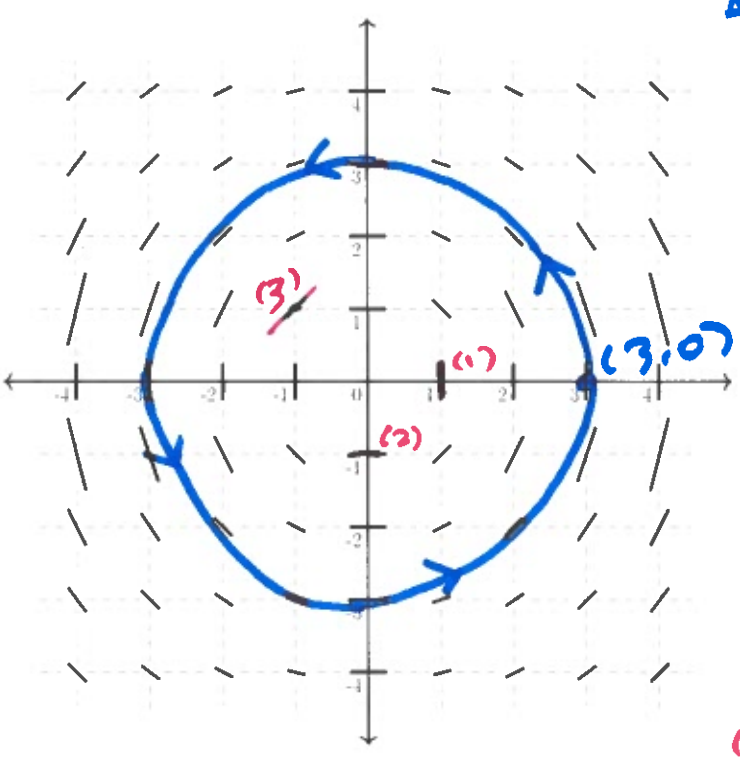
One attribute of a direction field is it allows the study of asymptotic behavior ($\lim_{x\to\infty} y(x)$) by purely a visual means without any algebraic understanding of the solution.

Example: In Lecture 2 we noted that $x^2 + y^2 = C$ is a general solution to the equation

circles

$$\frac{dy}{dx} = -\frac{x}{y}$$

below is a direction field for the equation. If you follow the flow by starting at a certain coordinate, you can get a particular solution which designates a value for C, which solves the corresponding IVP.



At $(3,0)$ $x=3$
$y=0$

$3^2+0^2=C \Rightarrow C=9$

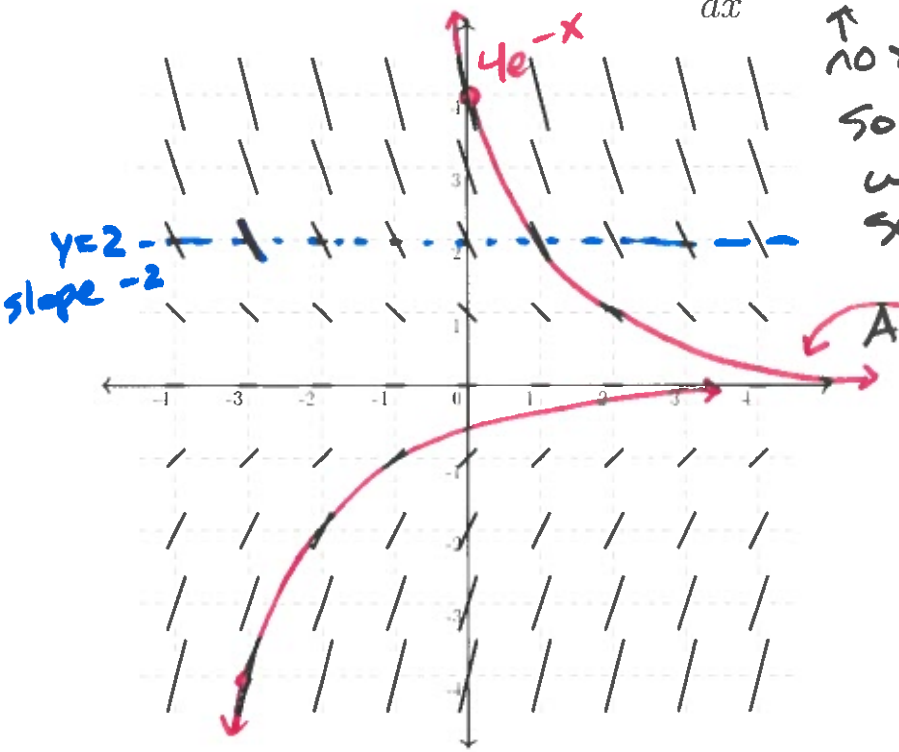
$x^2+y^2=9$ radius 3

Check a few values

	x	y	$\frac{dy}{dx} = -\frac{x}{y}$
(1)	1	0	undefined (vertical)
	0	1	0
	-1	0	undefined vertical
(2)	0	-1	0
	1	1	-1
(3)	-1	1	1
	1	-1	1
	-1	-1	-1

Example: Consider the equation $\frac{dy}{dx} = -y$.

no x's in $f(x,y) = -y$
so all coordinates with fixed y have same slope (i.e. on $y=a$)



A particular solution at $(0,4)$ is drawn

Can you make a claim as to the general solution? $y(x) = \underbrace{Ce^{-x}}_{\substack{\to 0 \\ x\to\infty}}$

At $(0,4)$ $\quad 4 = Ce^{-0} \Rightarrow C = 4$

$x=0, y=4$ $\quad y = 4e^{-x}$

Definition: A differential equation of the form $\frac{dy}{dx} = f(y)$ in which the independent variable is missing from the right side is called **autonomous**.

It follows that, for any autonomous equation, all slopes corresponding to a coordinate with a fixed y value are equal. Therefore, if we have any solution to an autonomous equation, a horizontal shift is also a solution.

shift $y = Ce^{-x} \to Ce^{-(x-a)} = Ce^{-x+a} = Ce^{-x}e^{a} = De^{-x}$

For equations which are not autonomous, how can we plot a direction field without evaluating the expression at every coordinate?

Example

Solve $\frac{dy}{dx} = k = -y \Rightarrow y = -k$ horizontal lines. (autonomous)

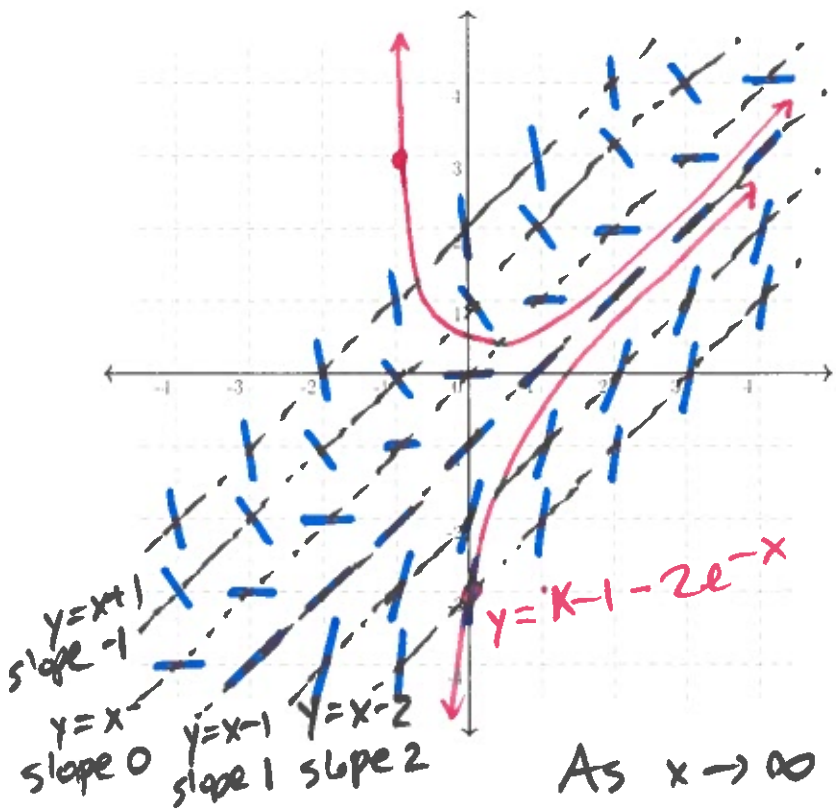
Definition: An **isocline** is a set of points in a direction field where all solutions have the same slope.

That is, if we fix $\frac{dy}{dx}$ equal to a constant, say k, we obtain an equation

$$\boxed{k = \frac{dy}{dx}} = f(x, y)$$

which is an implicit curve $f(x, y) = k$ that is an isocline. All of the coordinates on this curve in the direction field must have slope k. Thus, we can plot a direction field by determining its isoclines.

Example: Consider the equation $\frac{dy}{dx} = \overbrace{x - y}^{f(x,y)}$. Use isoclines to plot the direction field and determine the asymptotic behavior of all solutions.



Set $\frac{dy}{dx} = k \Rightarrow x - y = k$

$\Rightarrow y = x - k$

lines of slope 1.

k	$y = x - k$ or $f(x,y) = k$	slope
0	$y = x - 0 \Rightarrow y = x$	0
1	$y = x - 1$	1
2	$y = x - 2$	2
3	$y = x - 3$	3
-1	$y = x + 1$	-1

As $x \to \infty$ $y \to x - 1$

or solutions approach

$y = x - 1$

Note: General Solution is $y = \underbrace{x-1}_{} + \underbrace{Ce^{-x}}_{\to 0 \ x \to \infty}$

Example: Use a plotter or computer algebra program such as GeoGebra (`https://www.geogebra.org/m/W7dAdgqc` is a link to a Geogebra Direction Field Plotter) to sketch the direction fields for

1. $\dfrac{dy}{dx} = \sin(x)\sin(y)$

2. $\dfrac{dy}{dx} = x^2 + y^2$

3. $\dfrac{dy}{dx} = 3 - y + \dfrac{1}{x}$ (What is the asymptotic behavior?)

   $y \to \approx 3$

4. $\dfrac{dy}{dx} = \dfrac{x}{y}$ (What is the particular solution for the curve through $(0, -1)$?)

   hyperbolas $y^2 - x^2 = 1$ $\quad x \quad y$

   $(-1)^2 - 0^2 = 1$ ✓

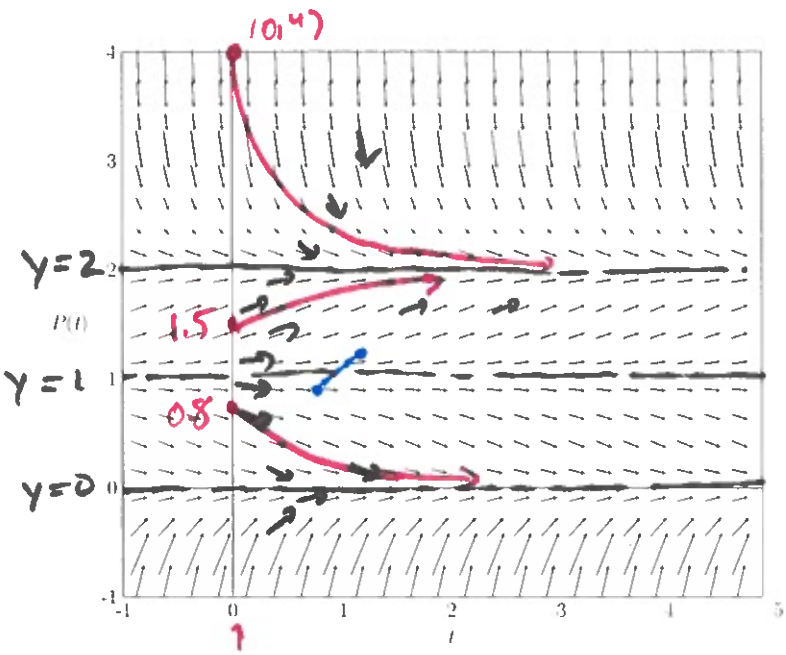
   particular solution is $\boxed{y^2 - x^2 = 1}$

Example: The equation

$$\frac{dP}{dt} = \overbrace{-P(P-1)(P-2)}^{\text{autonomous no } t\text{'s}}$$

0 1 2

is an example of a **logistic equation** where $P(t)$ is the population (in thousands). The direction field is given below:



(a) If the initial population is 4000, what can you say about $\lim_{t\to\infty} P(t)$?

2000

(b) If $P(0) = 1.5$, what is $\lim_{t\to\infty} P(t)$?] carrying capacity

2000

(c) If $P(0) = 0.8$, what is $\lim_{t\to\infty} P(t)$?

0 extinction

(d) Can a population of 900 ever increase to 1100?

No. must decrease towards 0.